

FREE STRENGTHEN

2023 Spring Semester Office Project

Design Studio 44 Prof. Won Suk Lee

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CONTEXT

BUILDING CODE

SITE ANALYSIS

DESIGN PROCESS

Concept
Mass Strategy
Program

PLAN

Site Plan
1F/Typical office/Shared space/Recharging floor/B1/Parking Plan
Section Plan
Elevation

DETAIL & STRUCTURE

Vertical Detail Section
Horizontal Detail Section
Structure Modeling
Exploded Axonometric
Ceiling Plan
Ventilation

RENDER

PHYSICAL MODEL

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Building Code

FAR(Floor Area Ratio)

Site area:784m²
FAR:600~800% or less
Total Floor Area=4704m²~6272m²

BCR(Building Coverage Ratio)

Site area:784m²
BCR: 60% or less
Building Area=~784*60%=~470.4m²

Building Height

Lower than 120m

Parking Lot

1 parking lot per 122m²
Handicap Parking: 3% of total parking lot should be planned

(Special) Escape Stairs

More than 5 upper floors / 2 underground floors -> escape stairs
+11 upper floors / +3 underground floors -> special escape stairs

Landscape Plan

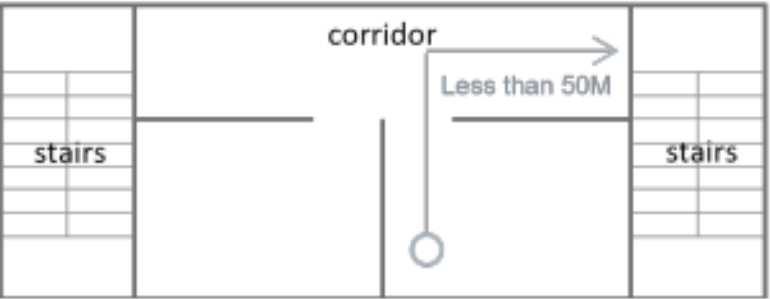
For buildings with a total floor area of 2000m² or more, an area of at least 15% of the land area shall be set as a landscaping area.
Landscape area=15%*784m²=117.6m²
The area shall not be less than 10m², and not less than 1m for its width.
Only half of the are for the green roof or green wall could be regarded as landscape area.

Public Open Space

If the total floor area is more than 5000 m² and less than 10000m², 5% of the land area should be secured by Open Space.
Public Open Space Area=5%*784m²=39.2m²

Direct Stairs

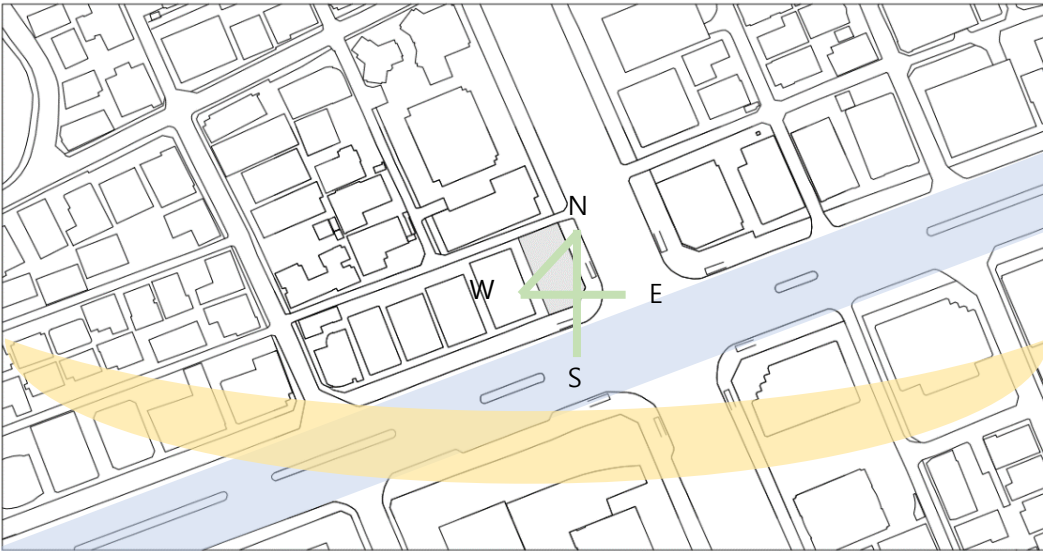
Reach stairs from everywhere within 50m(with sprinklers, 30m without sprinklers)
+3 floors and floor area over 400m² / underground floor over 200m²-> at least.2 stairs



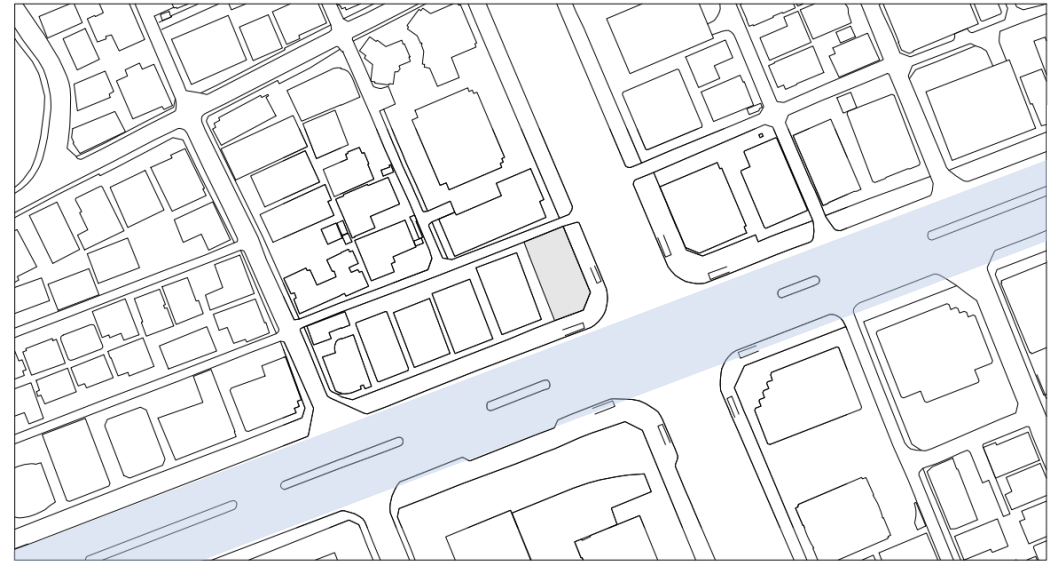
Elevator

Passenger elevator : 6th floor to top = +3000m² -> 1 elevator per 2000m² (-3000m²)
Emergency elevator : Building height +31m and largest floor area + 1500m² -> 1 elevator per 3000m² (-1500m²)

Site Analysis

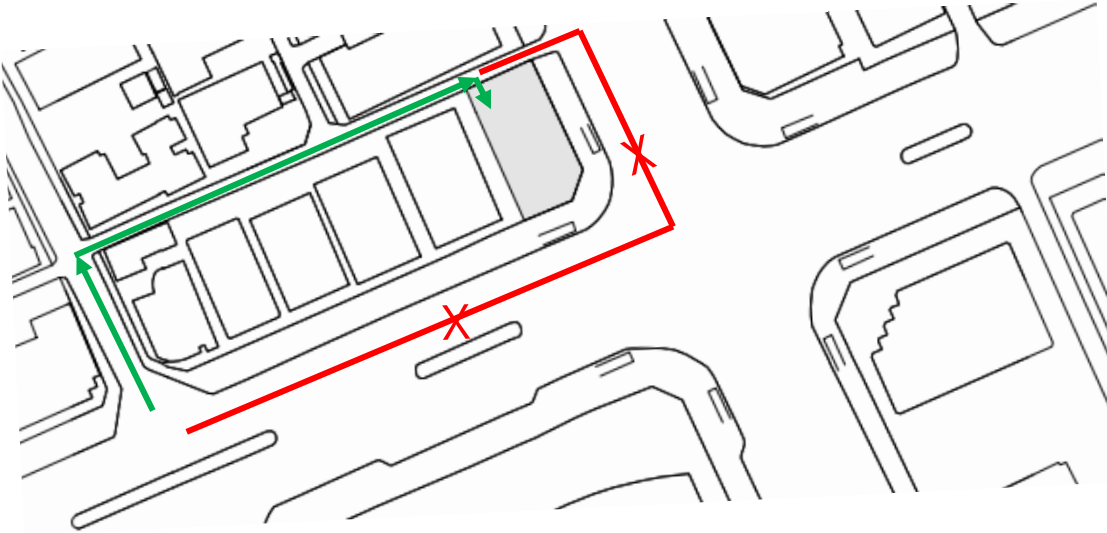


Sunlight



Noise

Site Analysis

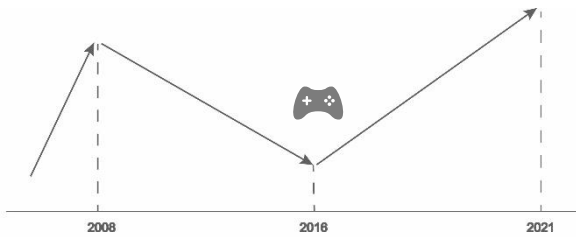


Vehicle Access

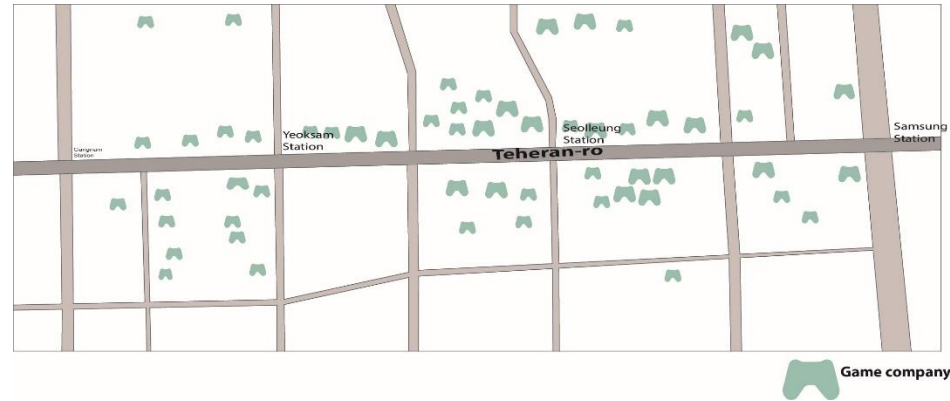


Transportation

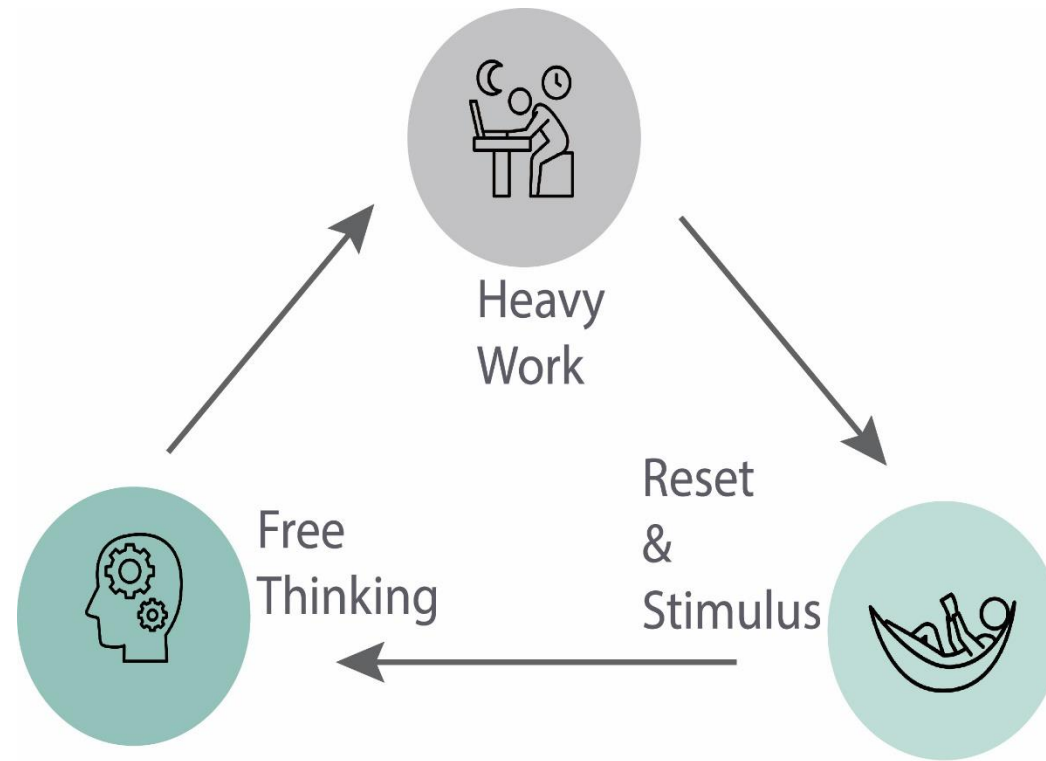
Design Process



Numbers of game company in Teheran-ro



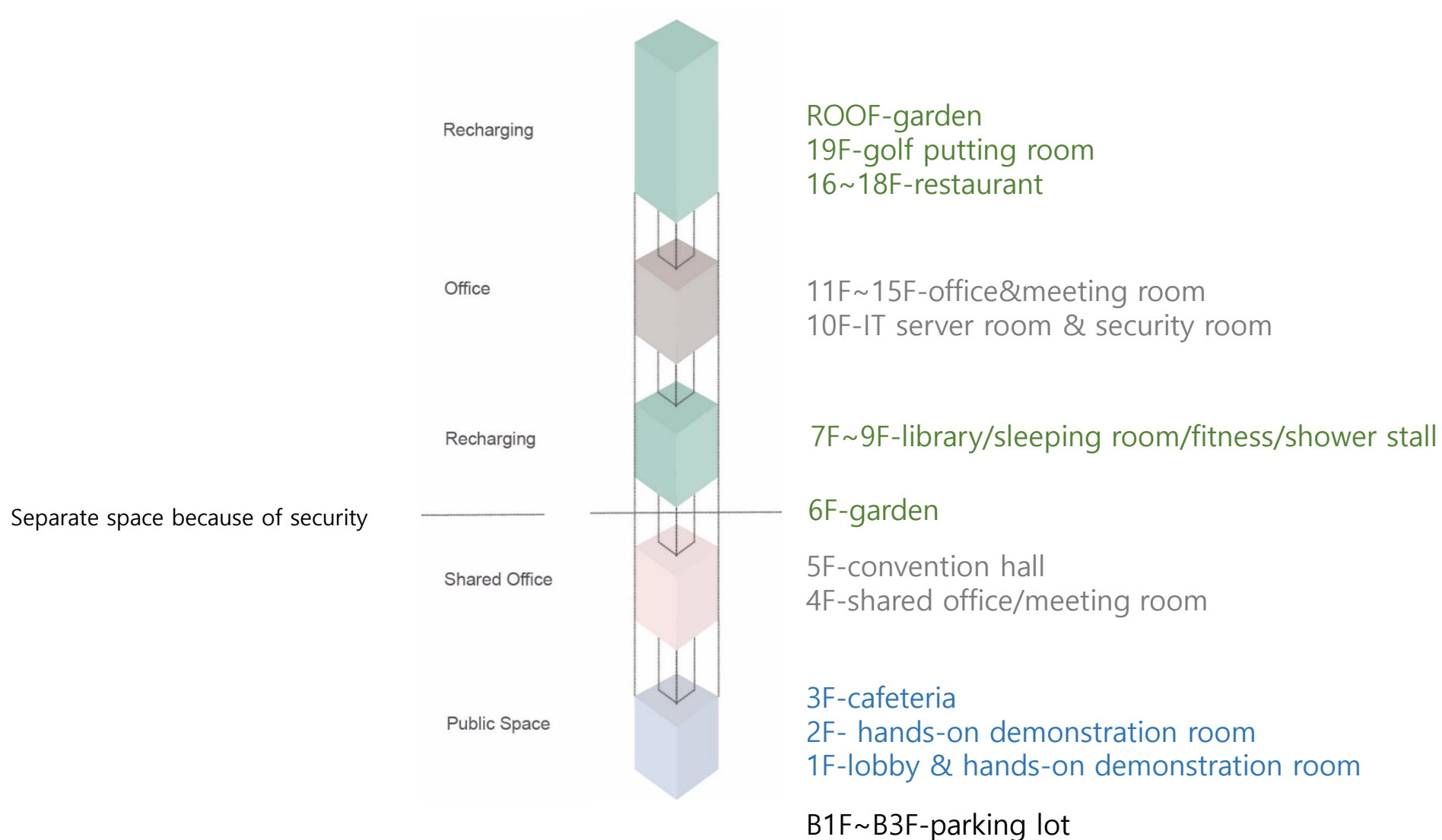
Design Process



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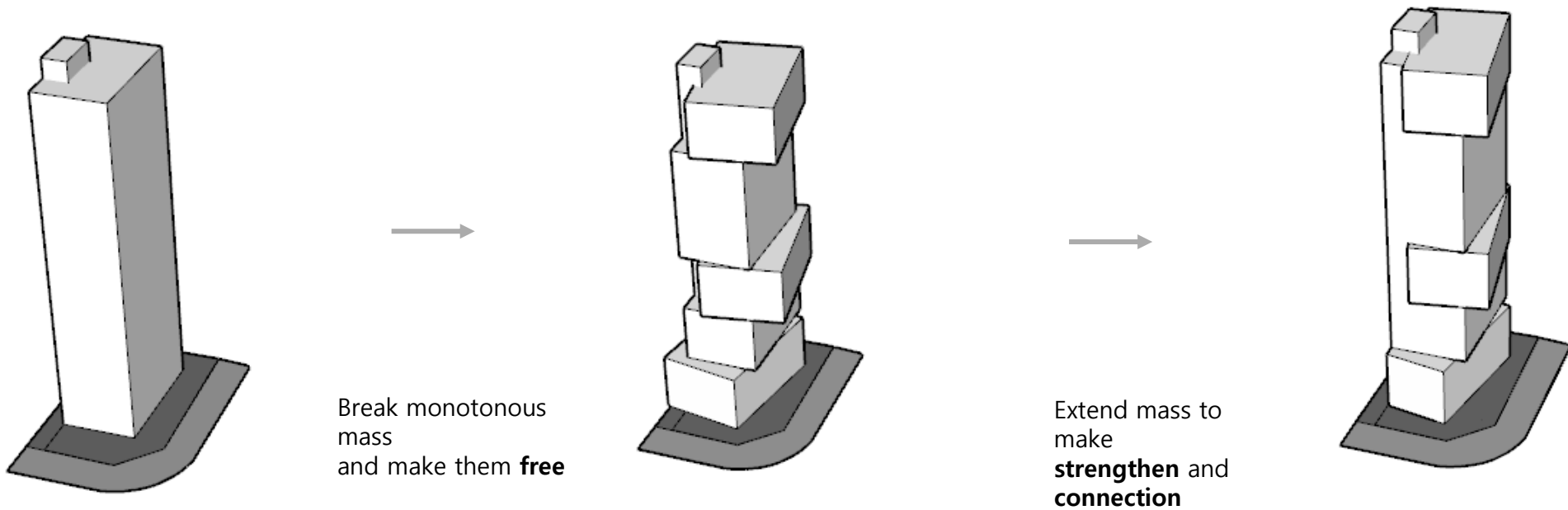
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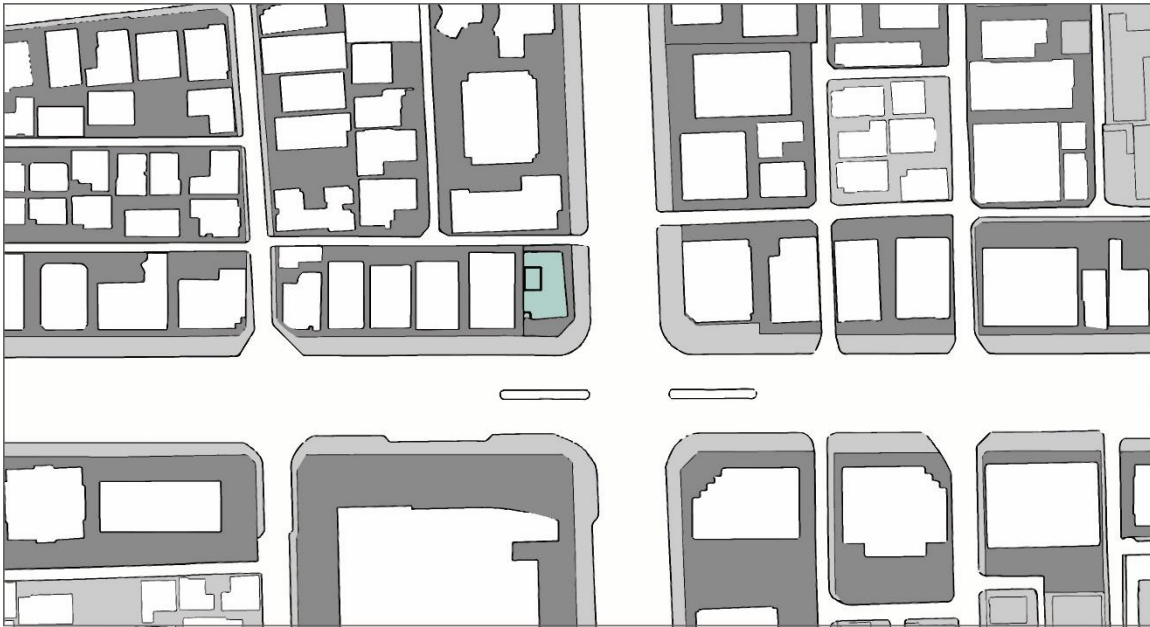
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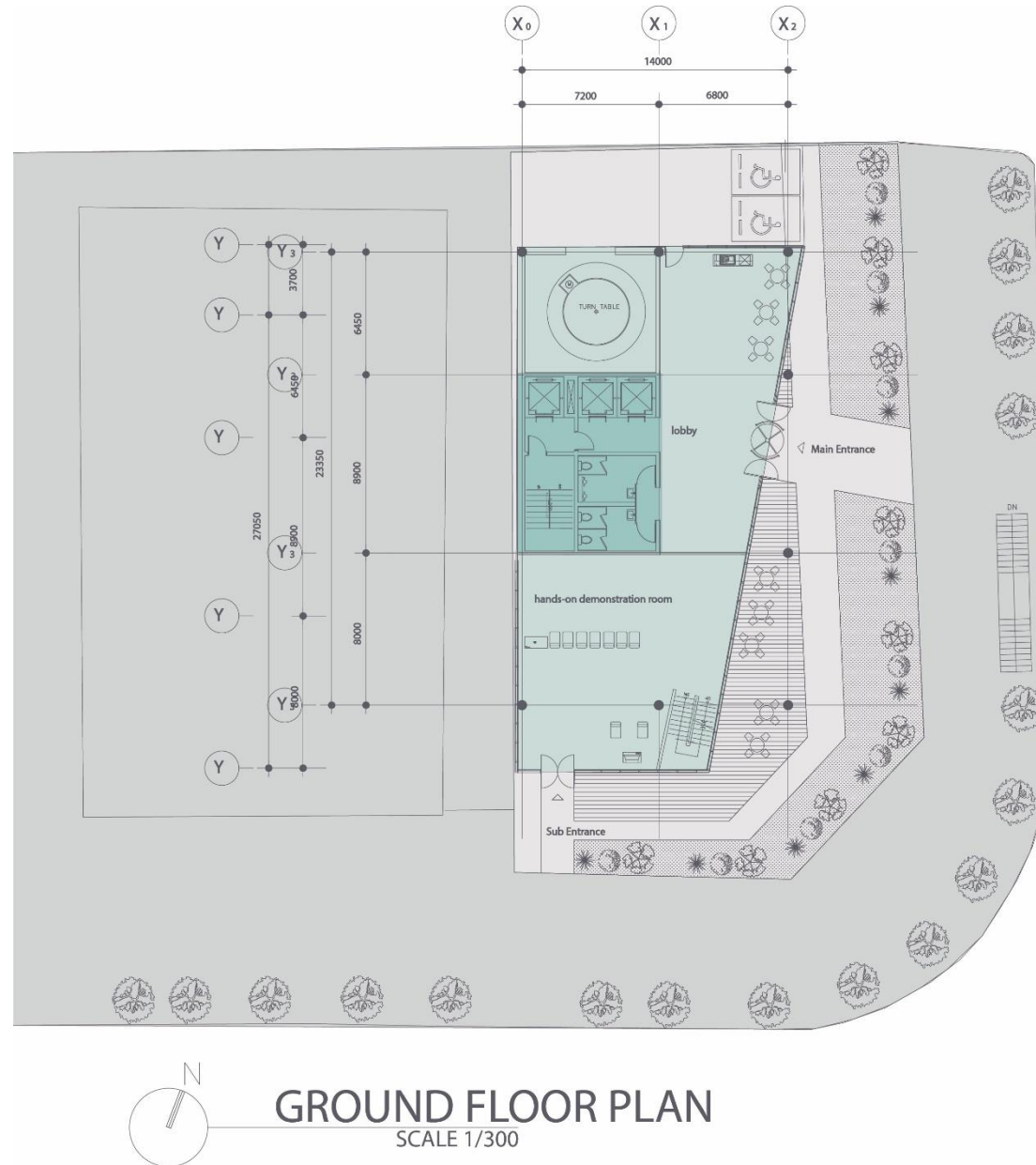
Plan

Site plan



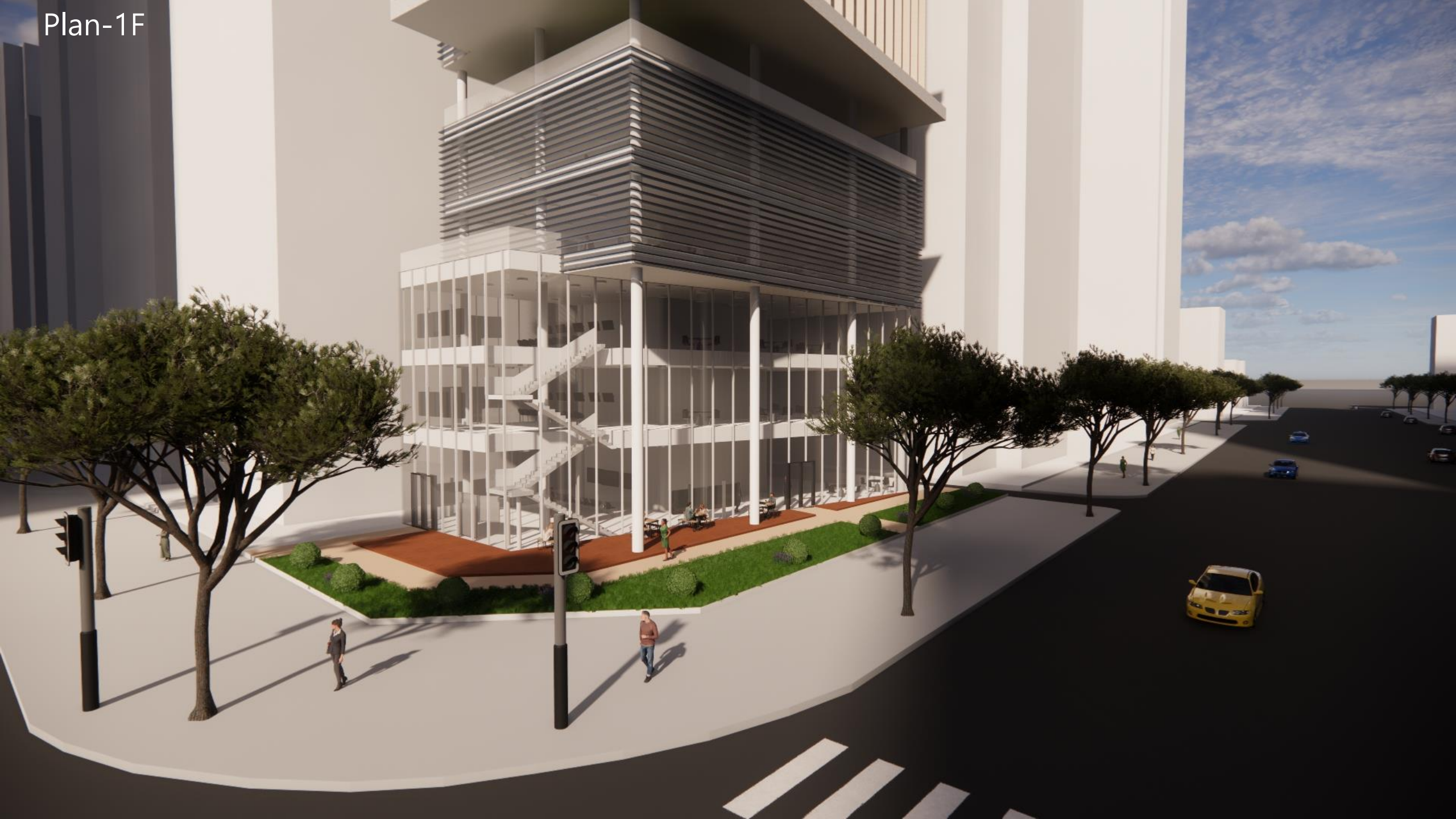
BCR:~411m²
FAR:~6200m²
Building Height: ~76m

Plan-1F

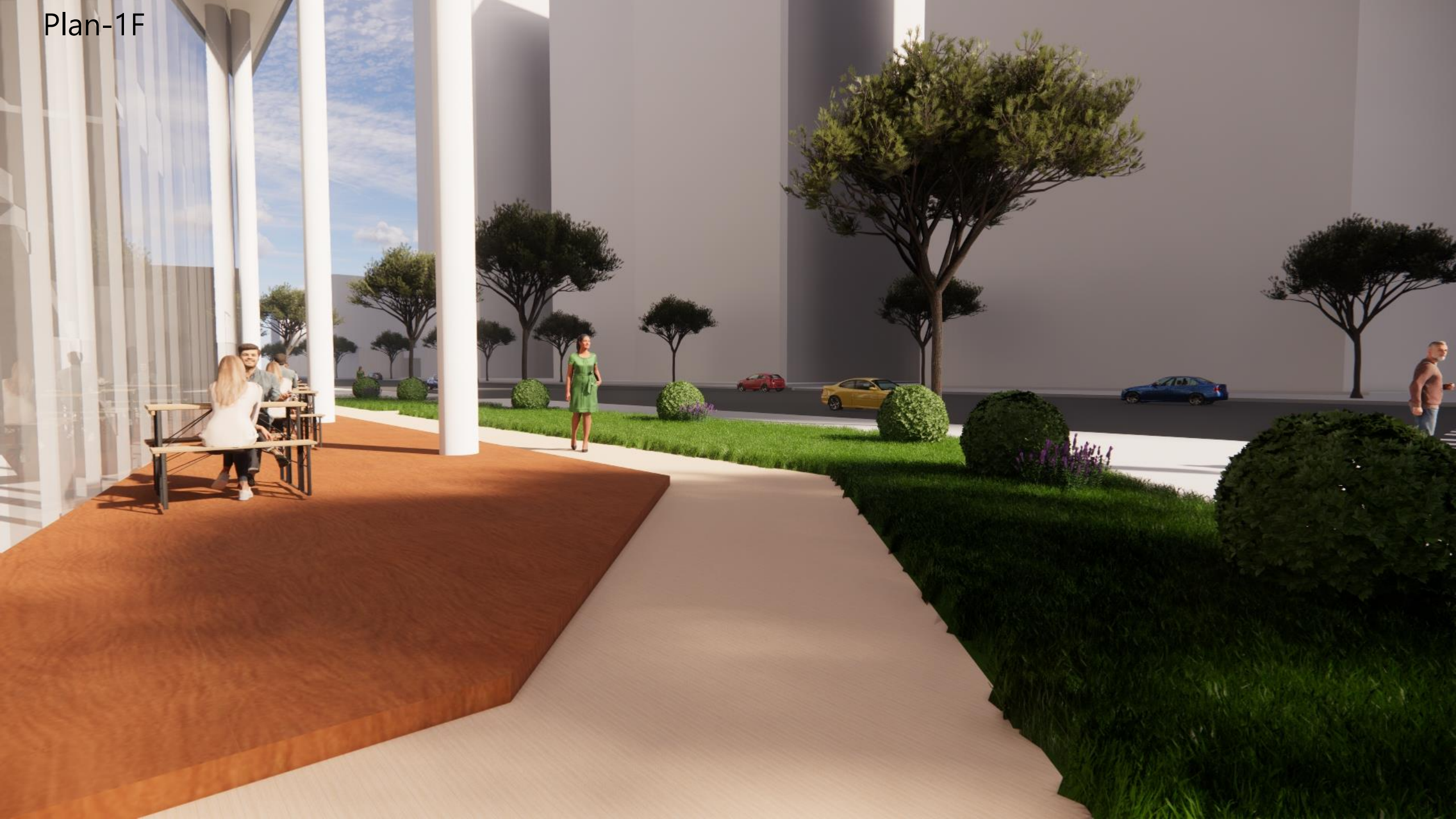




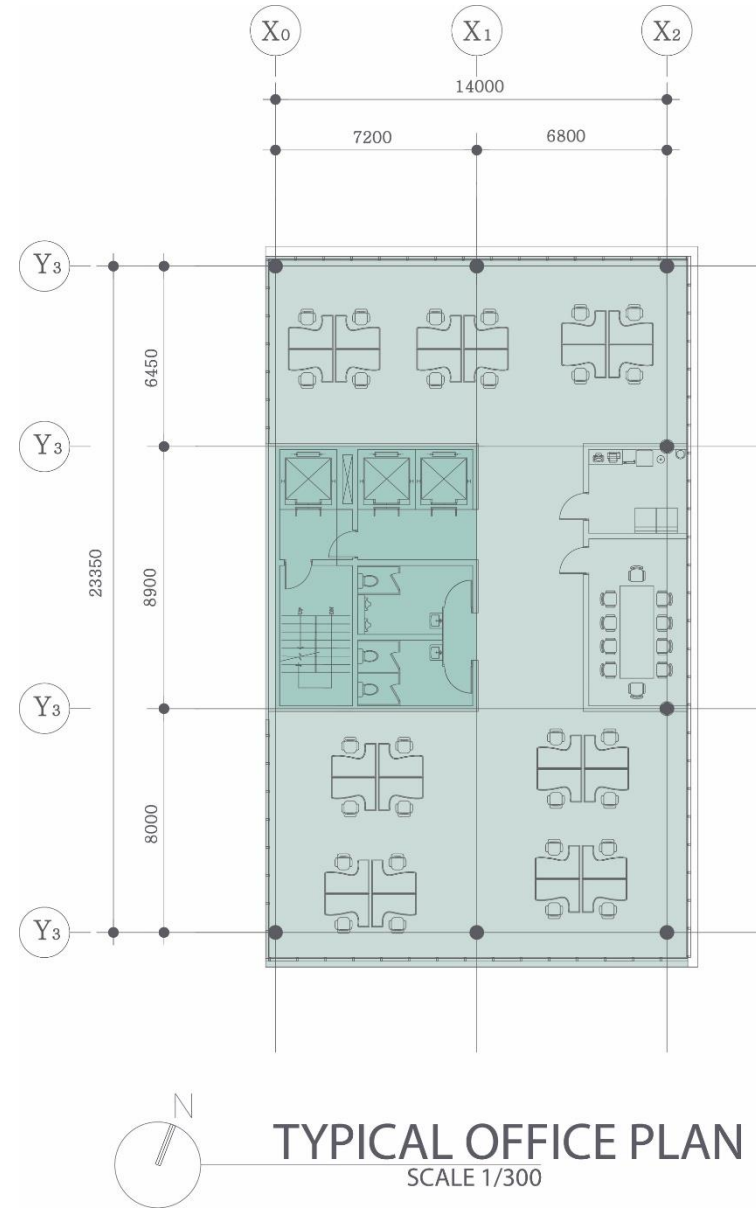
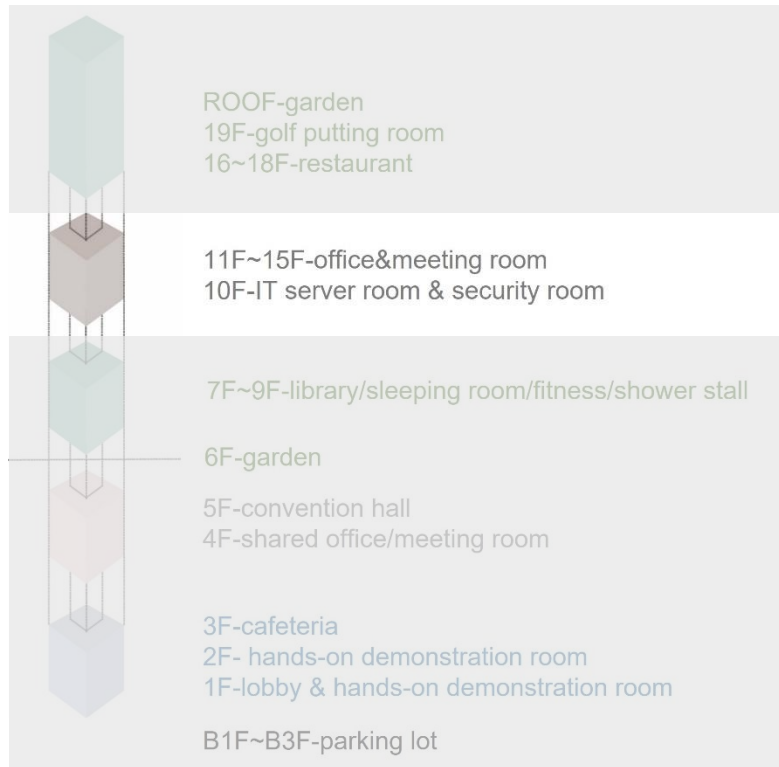
Plan-1F



Plan-1F



Plan-Typical Office



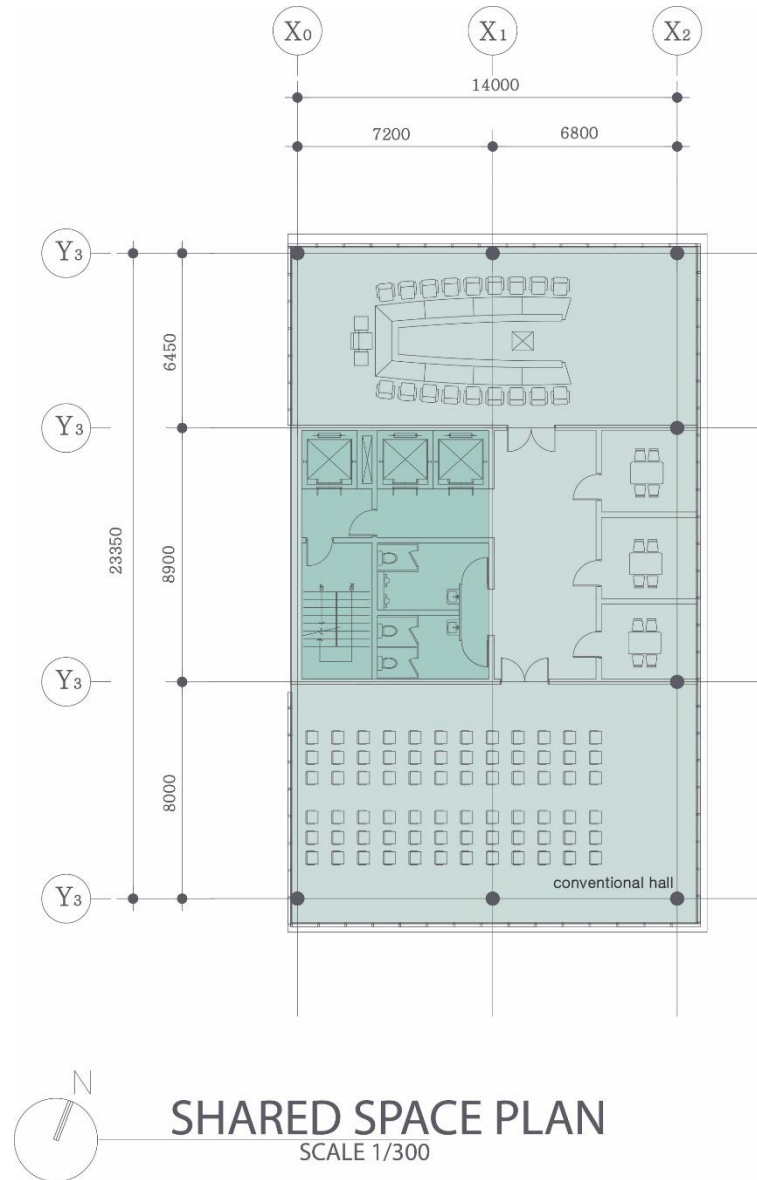
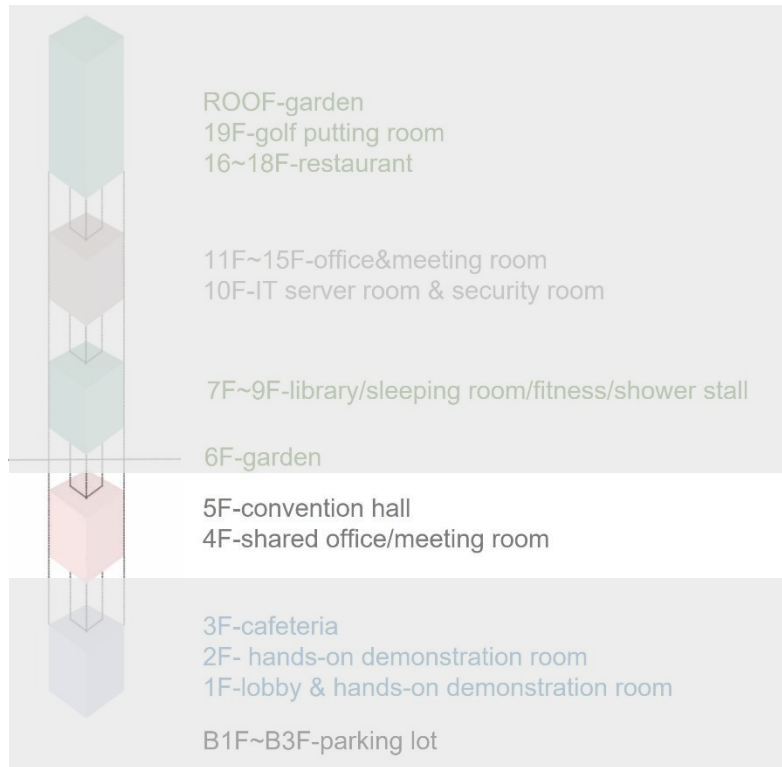
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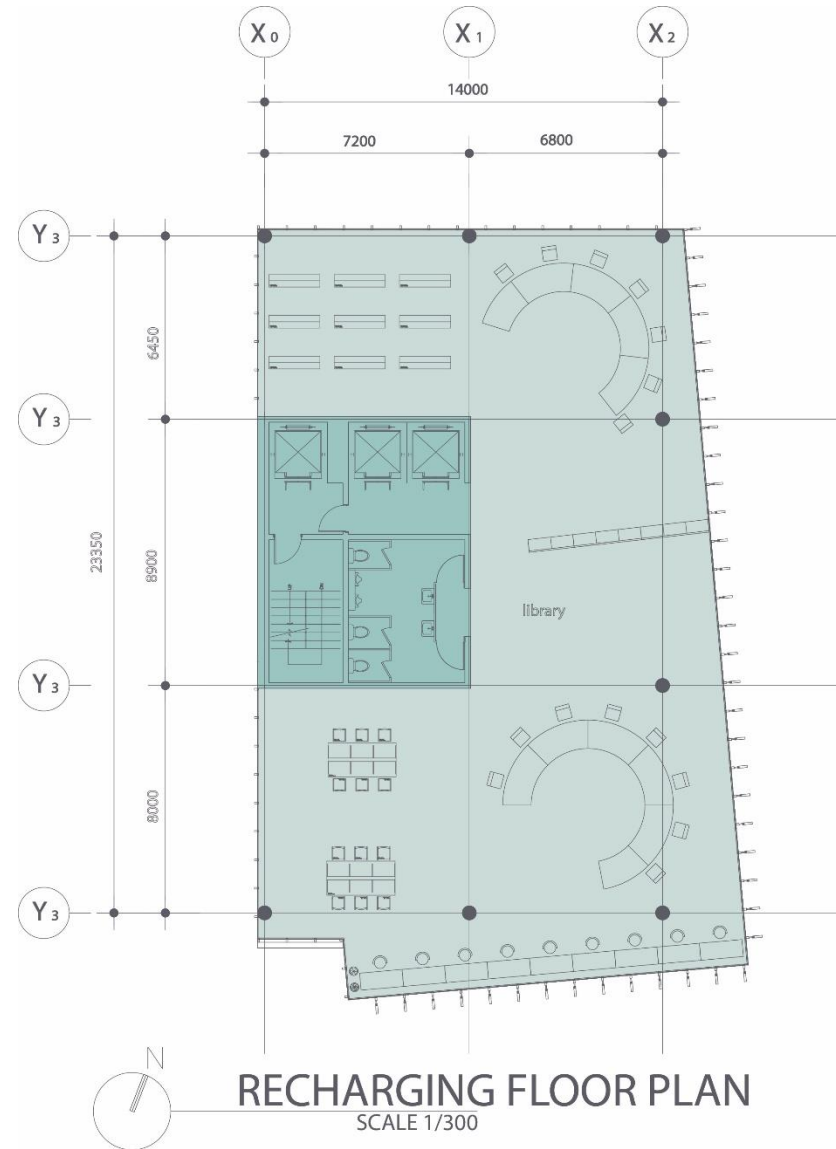
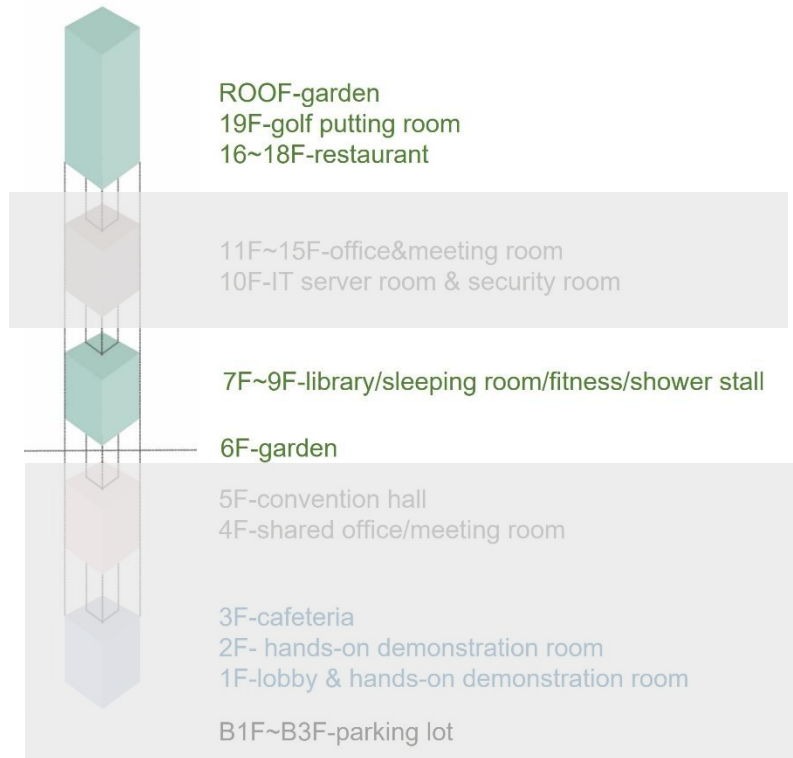
Plan-Typical Office



Plan-Shared Space



Plan-Recharging



Plan-Recharging



Plan-Recharging



Plan-Recharging



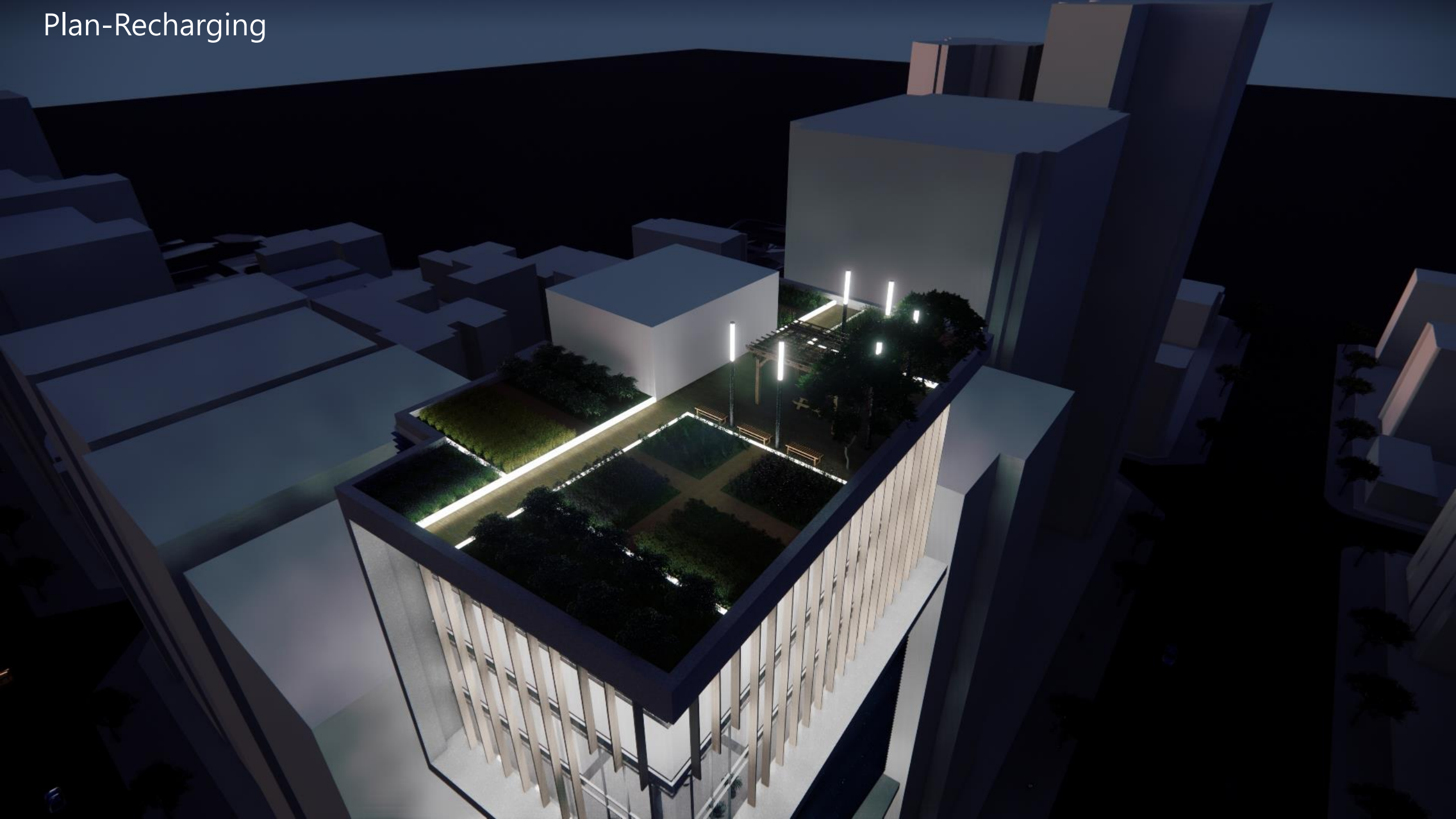
Plan-Recharging



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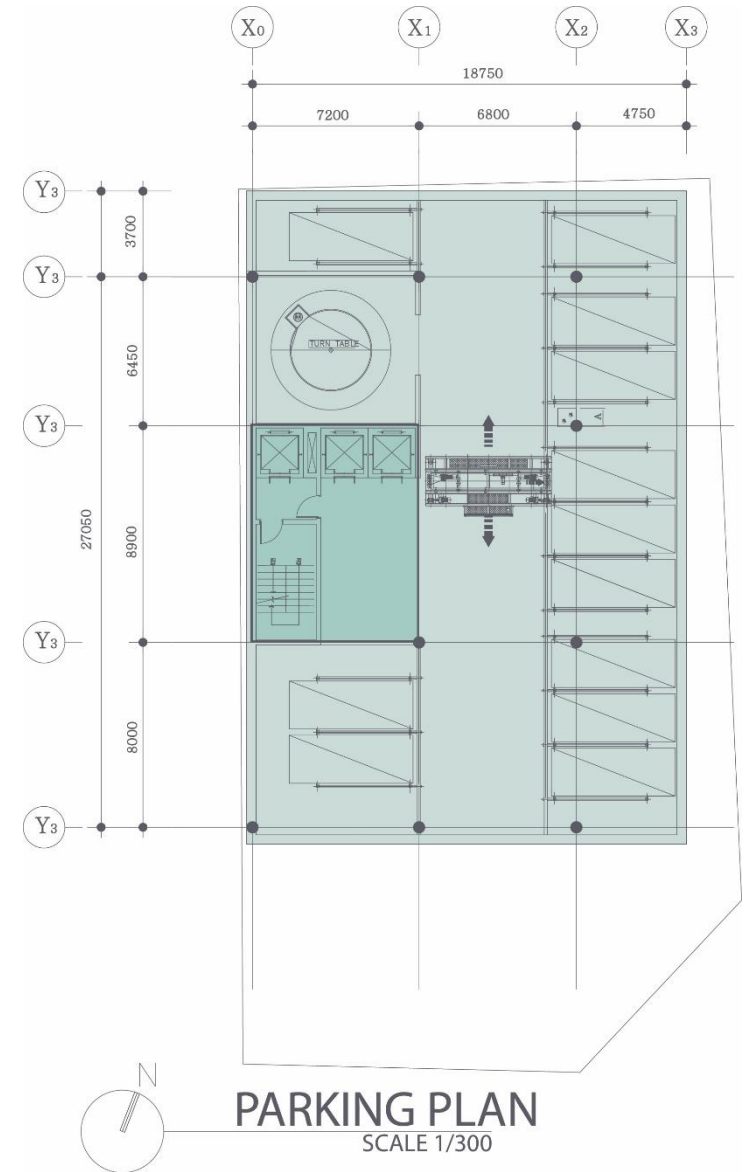
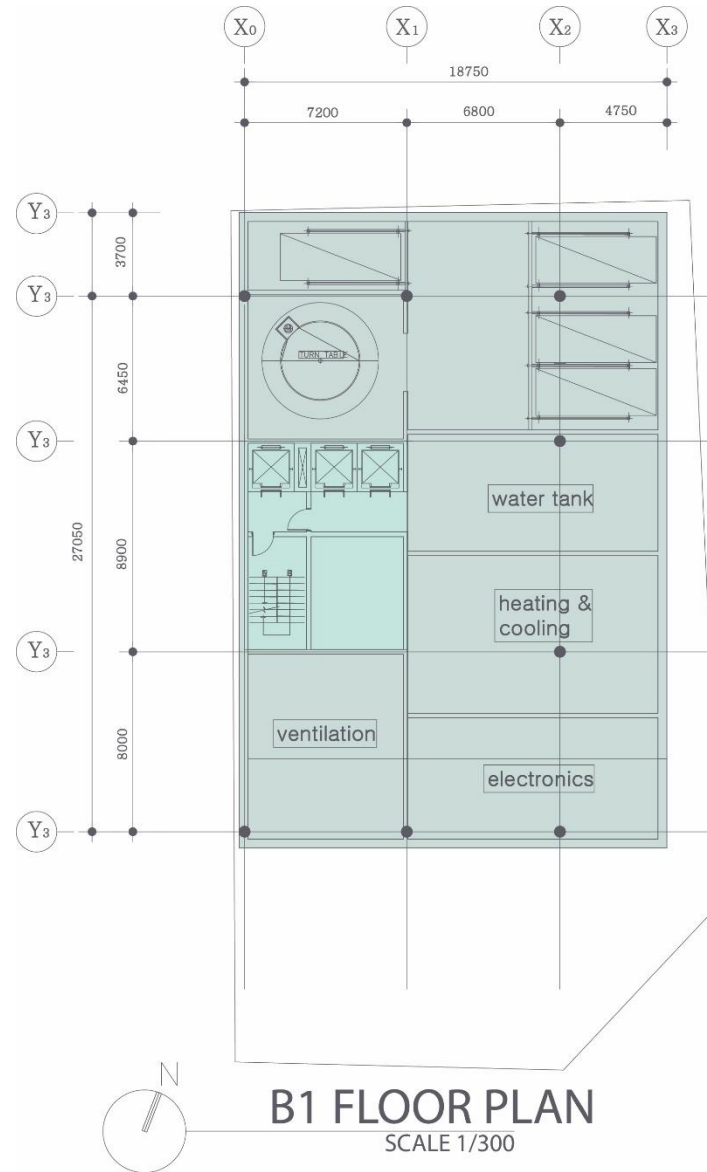
Plan-Recharging



Plan-B1 & Parking

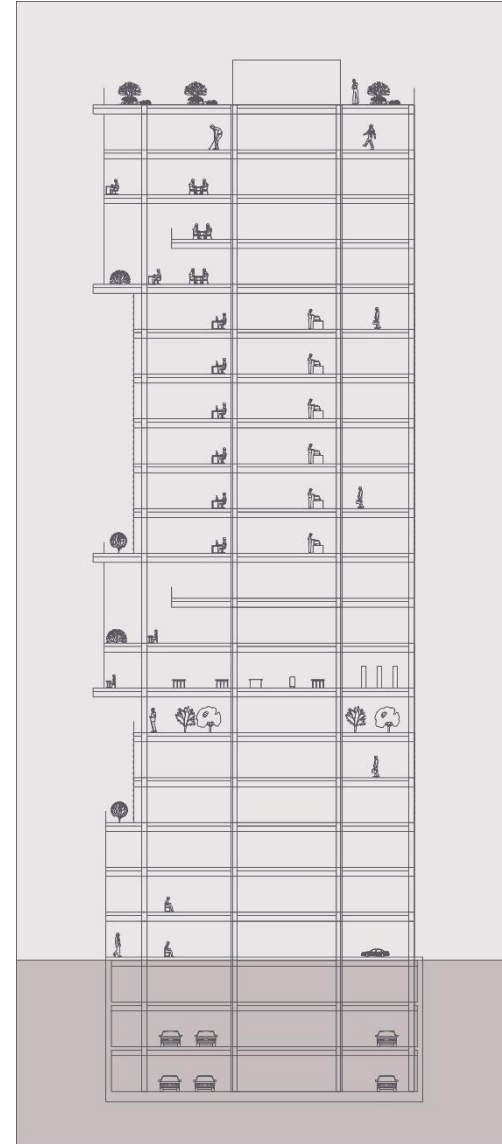
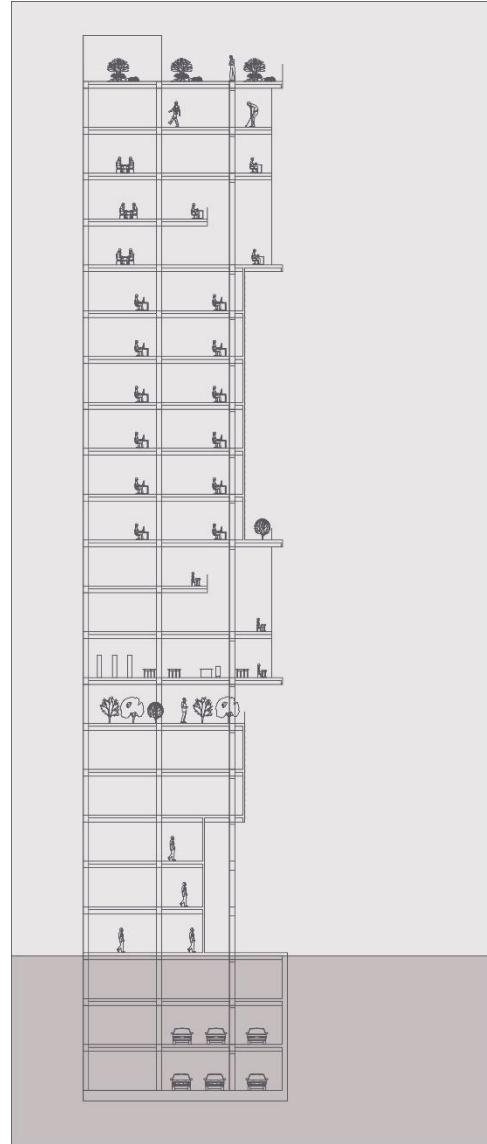


B1F~B3F-parking lot



Plan

Section



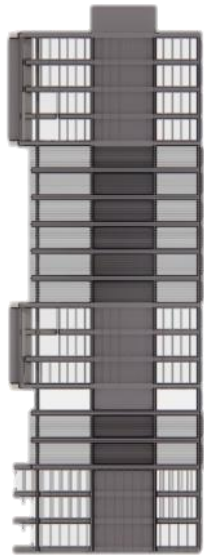
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Plan

Elevation



South elevation



East elevation



North elevation



West elevation

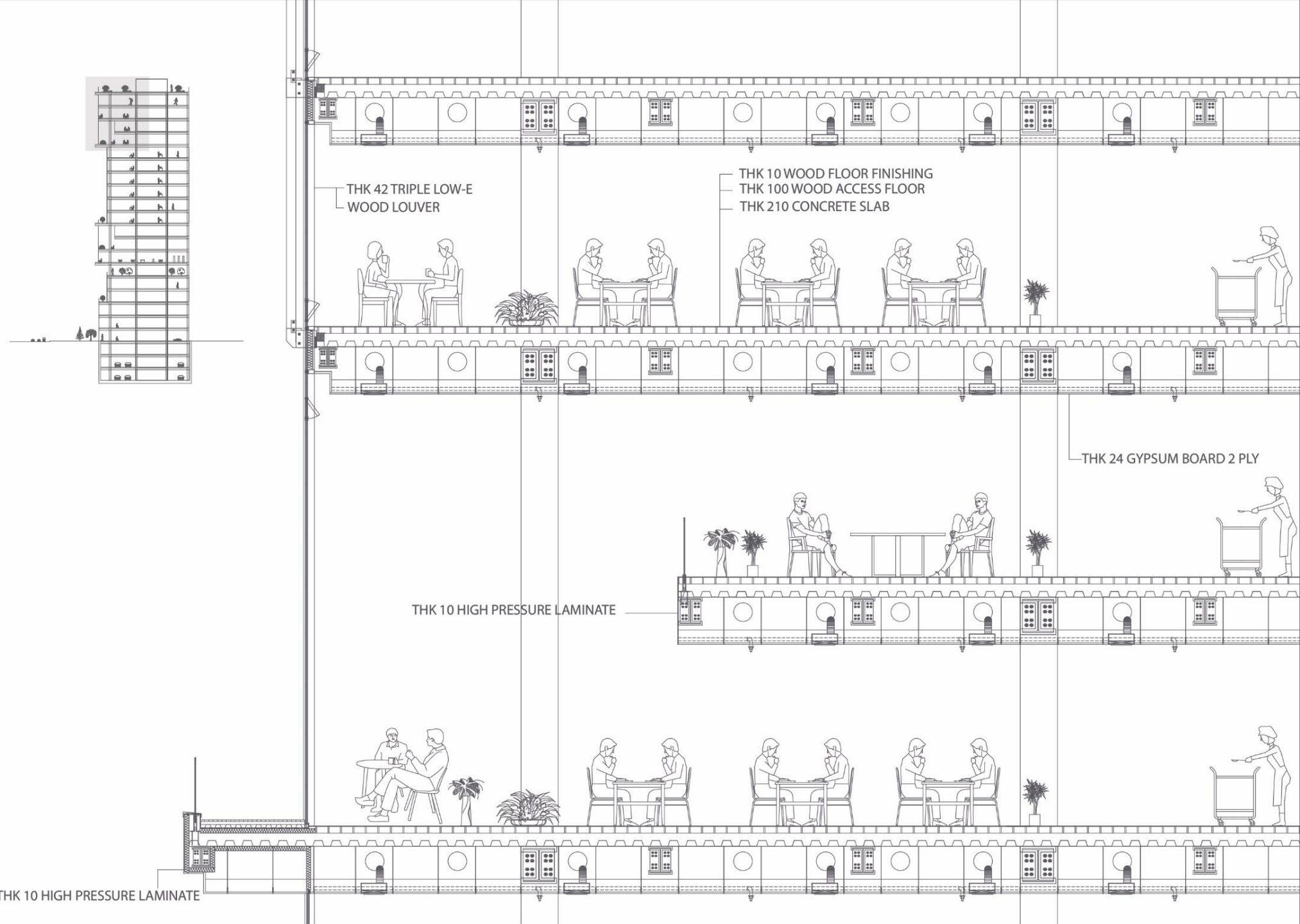
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Detail & Structure

1/50 Vertical Section



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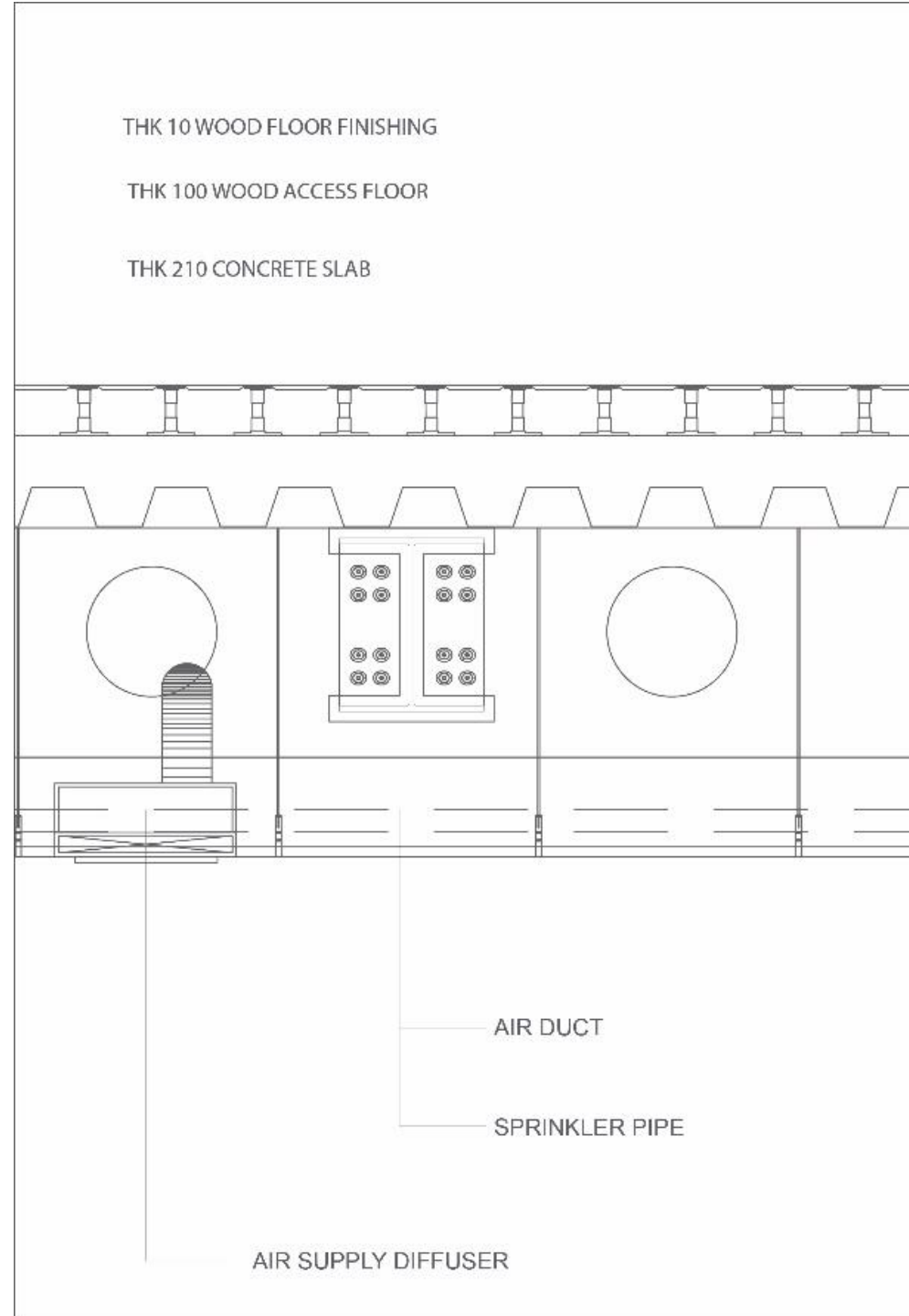
Detail & Structure

1/25 Detail Section

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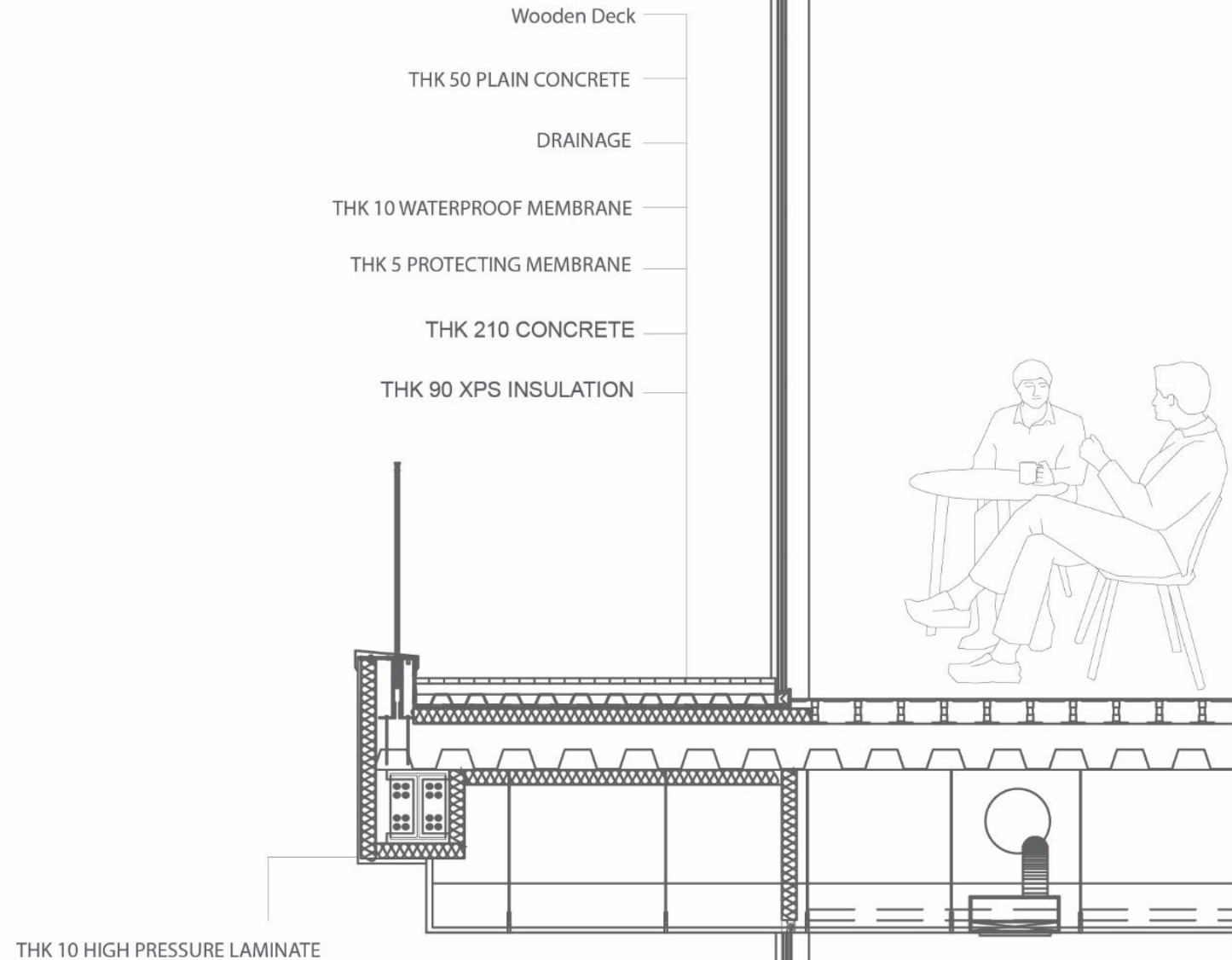
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1/25 Detail Section

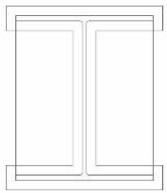
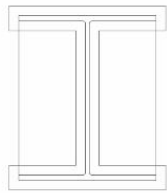
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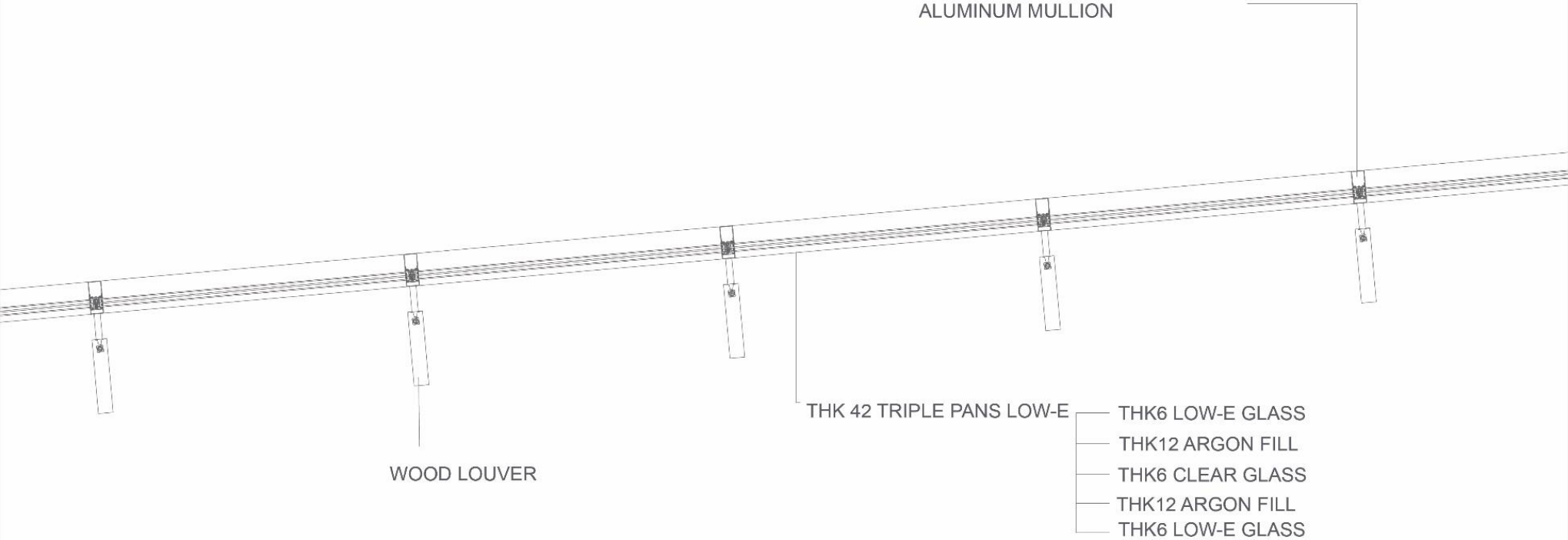


Horizontal Section

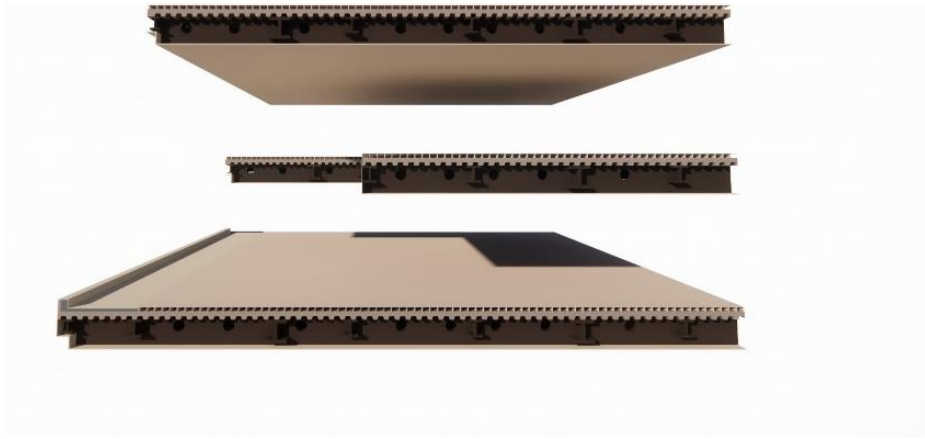
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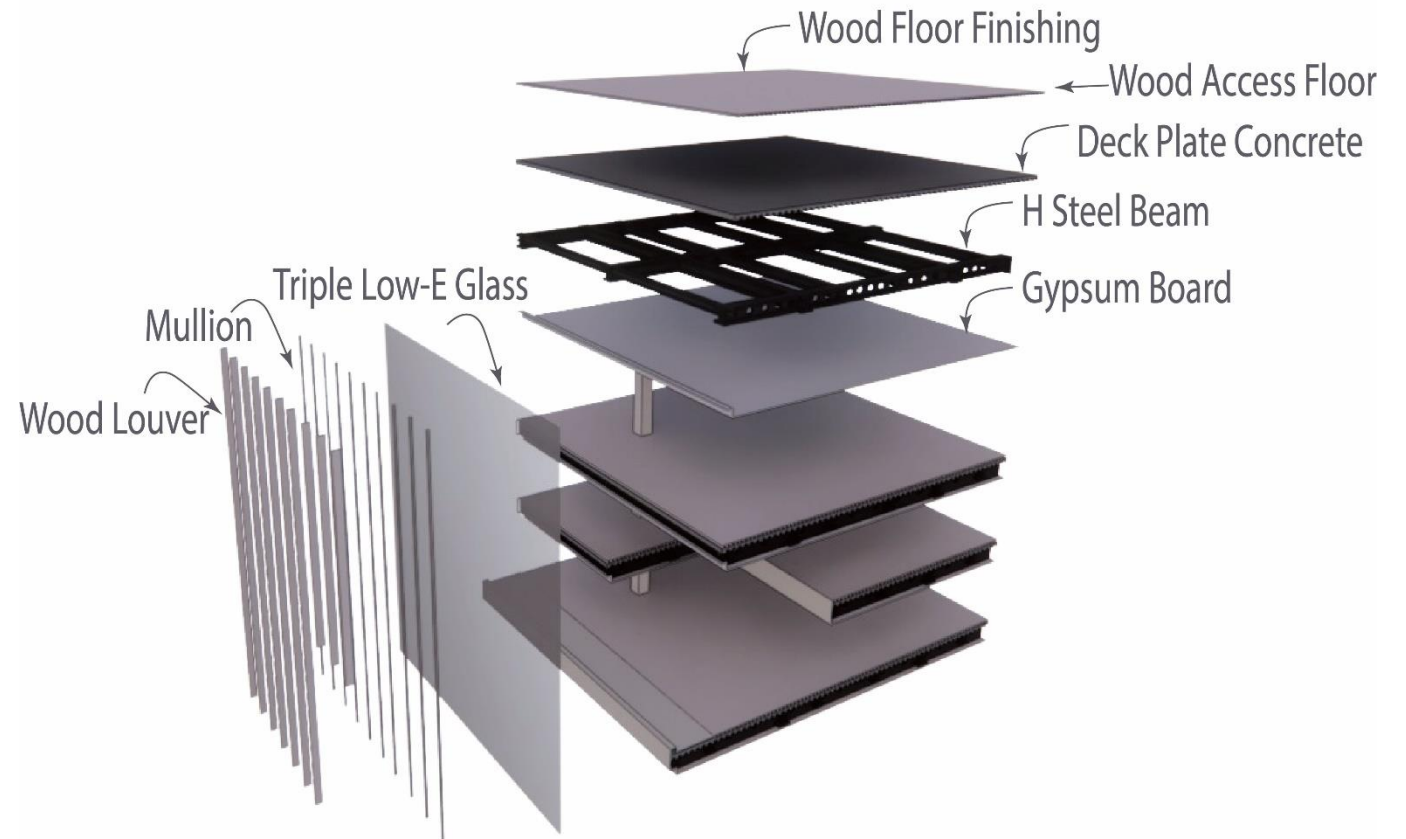
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Detail & Structure



Structure Modeling

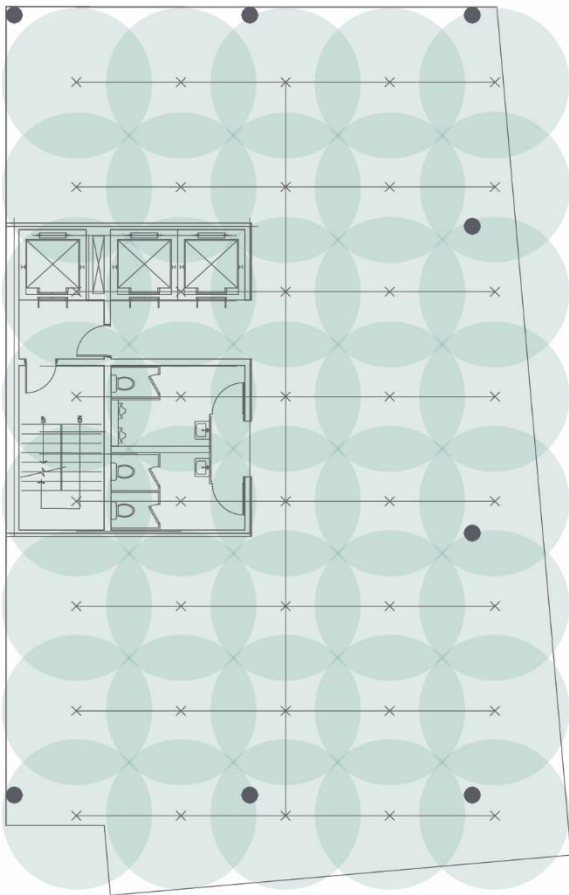


Exploded Axonometric

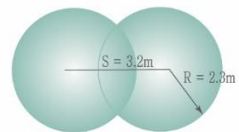
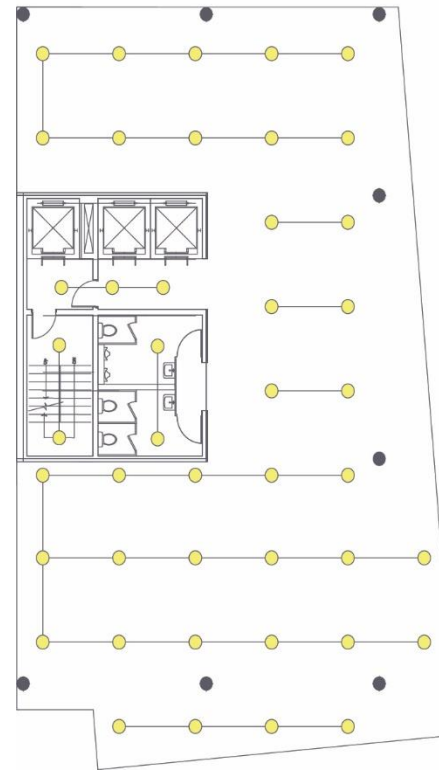
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Detail & Structure

Sprinkler Plan



Light Plan



S : Space between sprinklers
R : Radius of Distribution



3m

Ceiling Plan

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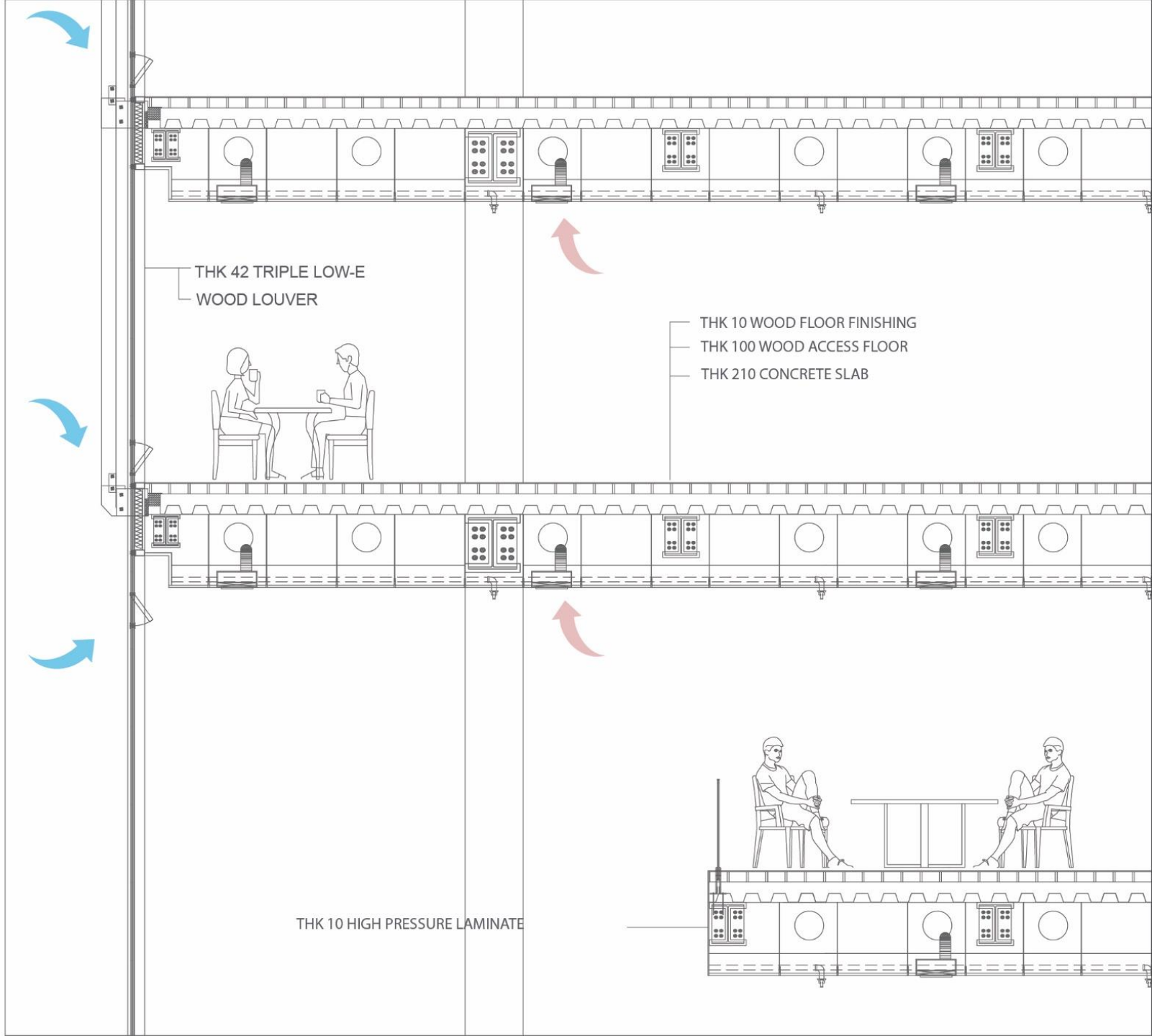
Detail & Structure

Ventilation

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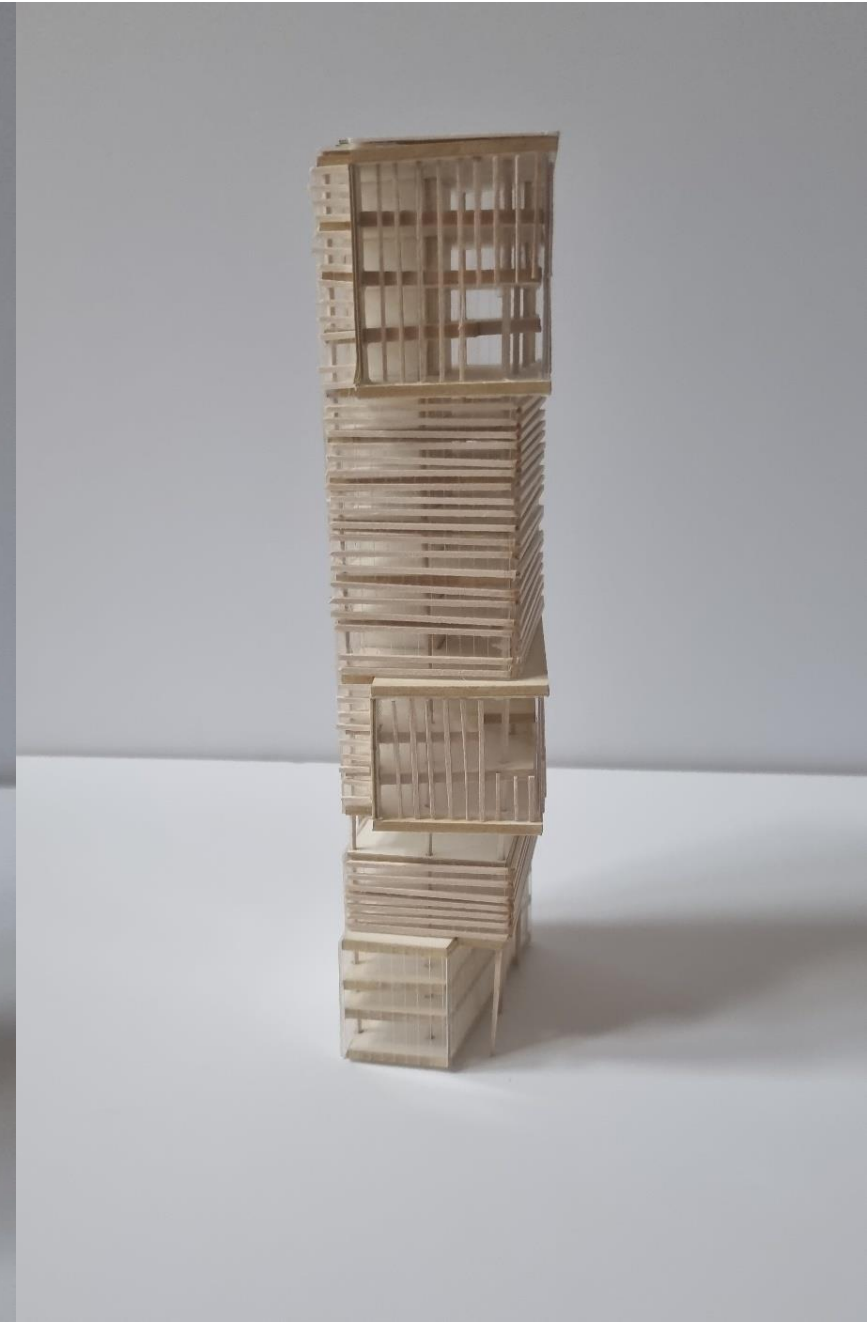
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Render



Physical Model



Physical Model



Physical Model

